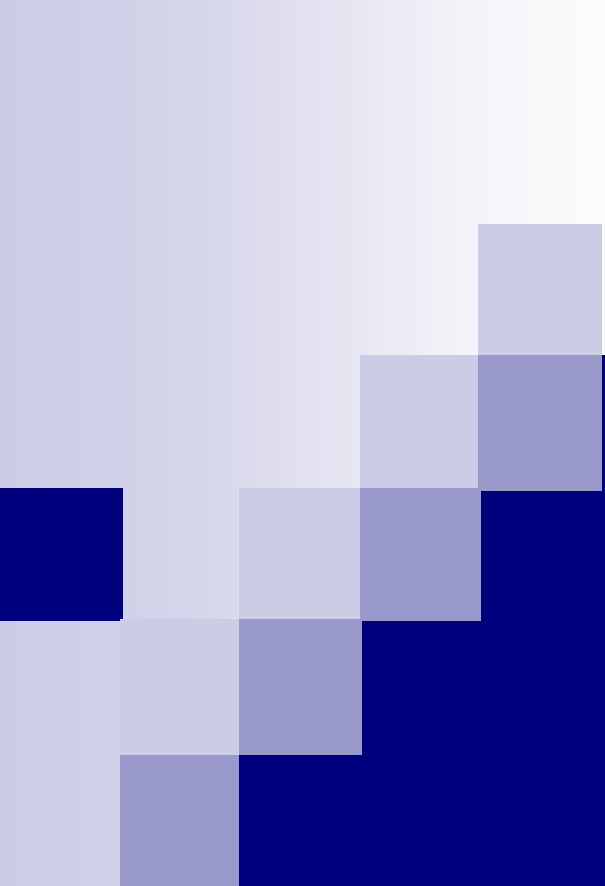




Software Configuration Management – 2 – Change Control Process

Lecture # 34

Recap and Today's Lecture

- Let us continue our discussion on the functions of software configuration management



SCM Functions

- Identification
- Version control
- Change control
- Configuration auditing
- Status accounting/reporting



SCM Function: Identification

- To control and manage software configuration items, each item must be separately named or numbered
- The identification scheme is documented in the software configuration management plan
- The unique identification of each SCI helps in organization and retrieval of configuration items

SCM Function: Version Control - 1

- Version control combines procedures and tools to manage different versions of configuration items that are created during the software process
- The naming scheme for SCIs should incorporate the version number

SCM Function: Version Control - 2

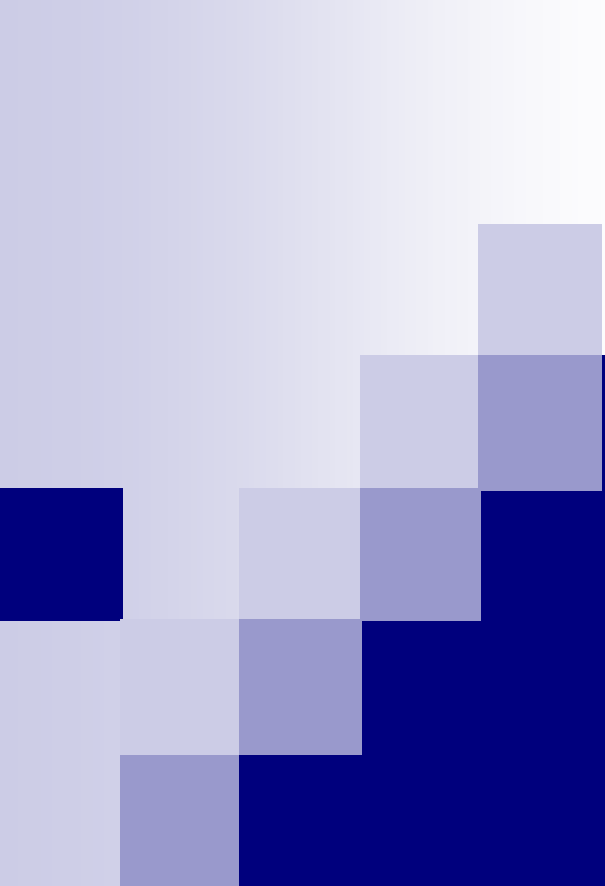
- Configuration management allows a user to specify alternative configurations of the software system through the selection of appropriate versions
- This is supported by associating attributes with each software version, and then allowing a configuration to be specified [and constructed] by describing the set of attributes

SCM Function: Change Control

- Change control is vital. Too much change control and we create problems. Too little, and we create other problems
- For large projects, uncontrolled change rapidly leads to chaos
- For medium to large projects, change control combines human procedures and automated tools to provide a mechanism for the control of change
- We'll talk about change control process after few minutes

SCM Function: Configuration Audit

- How can we ensure that the approved changes have been implemented?
 - Formal technical reviews/inspections
 - Software configuration audit
 - A software configuration audit complements the formal technical reviews/inspections by assessing a configuration item for characteristics that are generally not considered during review
 - The SCM audit is conducted by the quality assurance group



SCM Function: Configuration Audit Questions

SCM asks and answers the
following six questions

SCM Function: Audit Questions - 1

- Has the change specified in the ECO been made? Have any additional modifications been incorporated?
- Has a formal technical review/inspection been conducted to assess technical correctness?

SCM Function: Audit Questions - 2

- Has the software process been followed and have software engineering standards been properly applied?
- Has the change been “highlighted” in the SCI? Have the change date and change author been specified? Do the attributes of the configuration item reflect the change?

SCM Function: Audit Questions - 3

- Have SCM procedures for noting the change, recording it, and reporting it been followed?
- Have all related related SCIs been properly updated?

SCM Function: Status Accounting/Reporting - 1

- The status accounting function provides a corporate memory of project events that supports accomplishment of other configuration management items
 - What happened?
 - Who did it?
 - When did it happen?
 - What else will be affected?

SCM Function: Status Accounting/Reporting - 2

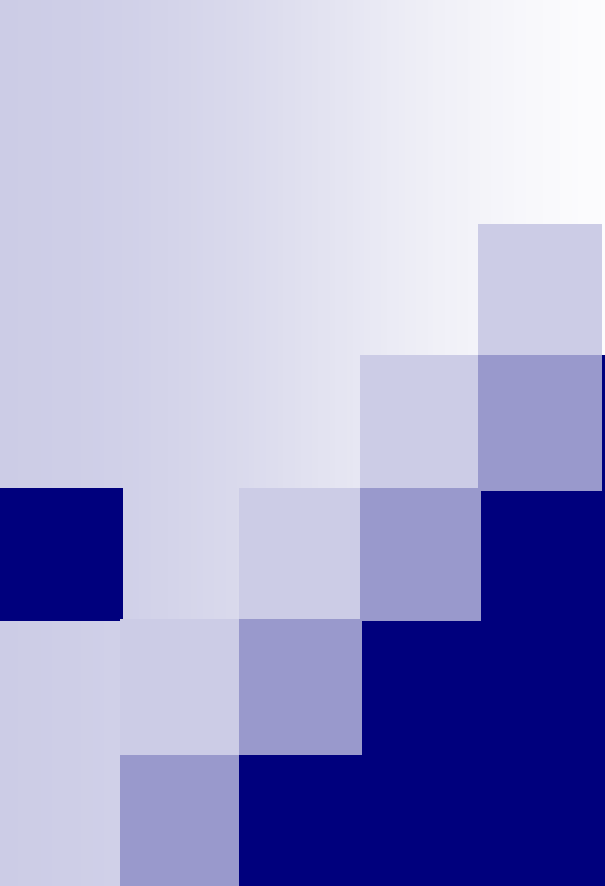
- Each time an SCI is assigned a new or updated identification, a configuration status reporting (or CSR) entry is made
- Each time a change is approved, a CSR entry is made
- Each time a configuration audit is conducted, the results are reported as part of CSR task
- Output from CSR may be placed in an on-line database for easy access, and CSR reports are issued on a regular basis to keep management and practitioners informed about important changes

SCM Function: Status Accounting/Reporting - 3

- Configuration status reporting plays a vital role in the success of a large software development project
- When many people are involved, it is likely that “the left hand not knowing what the right hand is doing” syndrome will occur
- Two developers may attempt to modify the same SCI with different and conflicting intents

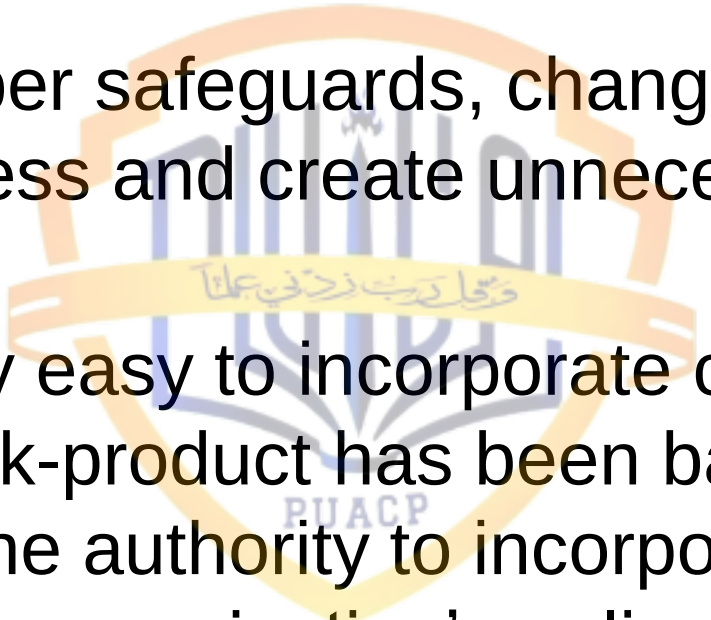
SCM Function: Status Accounting/Reporting - 4

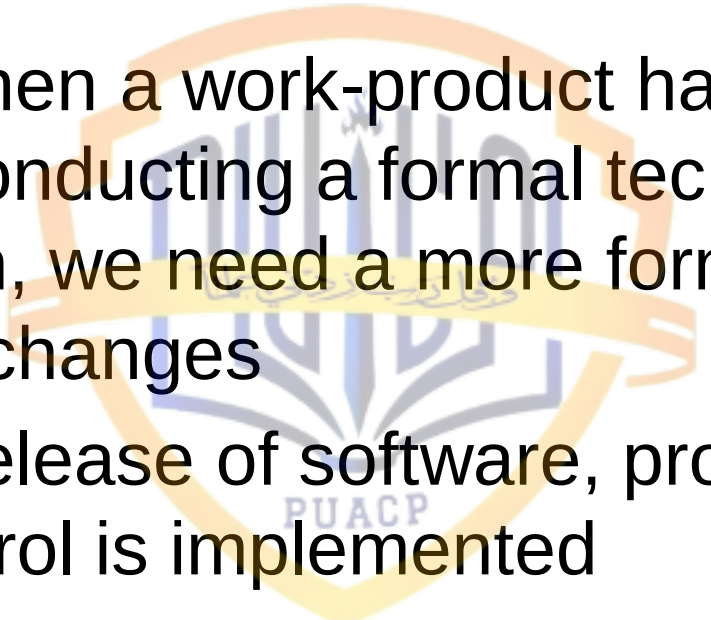
- A software engineering team may spend months of effort building software to an obsolete hardware specification
- A person who would recognize serious side effects for a proposed change is not aware that the change is being made
- Configuration status reporting helps to eliminate these problems by improving communication among all people involved

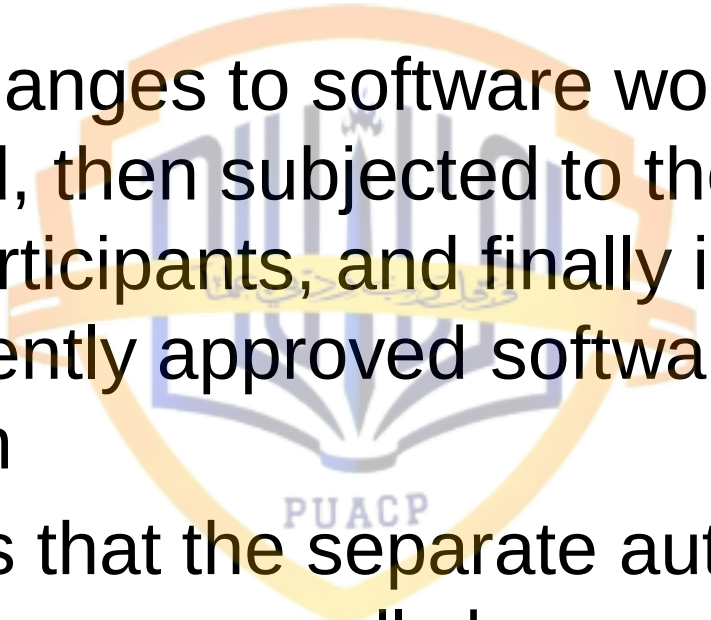


Change Control Process



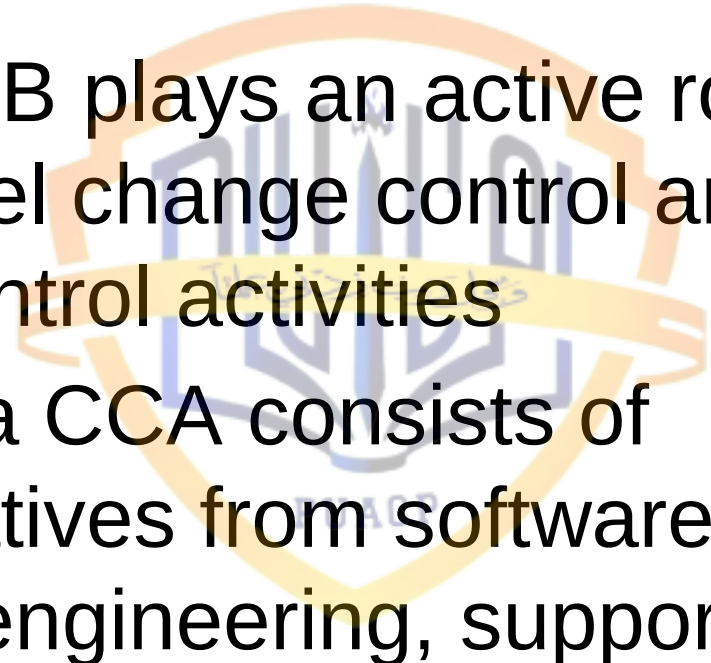
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- Without proper safeguards, change control can retard progress and create unnecessary red tape
 - It is relatively easy to incorporate changes before a work-product has been base-lined – the author has the authority to incorporate changes based on the organization's policy, project management's guidelines, and according to the technical needs of the project. We only need informal change control

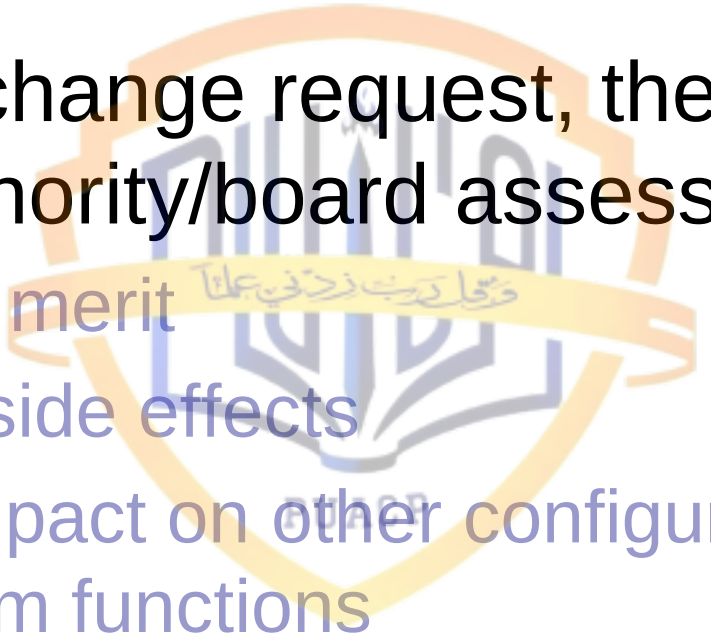
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- However, when a work-product has been base-lined after conducting a formal technical review or inspection, we need a more formal process to incorporate changes
 - Before the release of software, project level change control is implemented
 - However, when the software product is released to the customer, strict formal change control is instituted

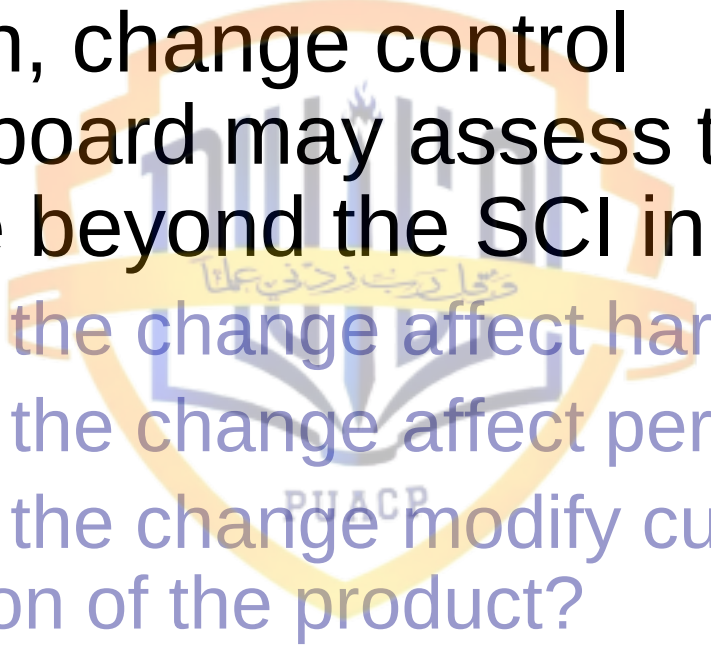
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- Proposed changes to software work-products are reviewed, then subjected to the agreement of project participants, and finally incorporated into the currently approved software configuration
 - This requires that the separate authority, which reviews and approves all change requests be established for every project from among the project participants

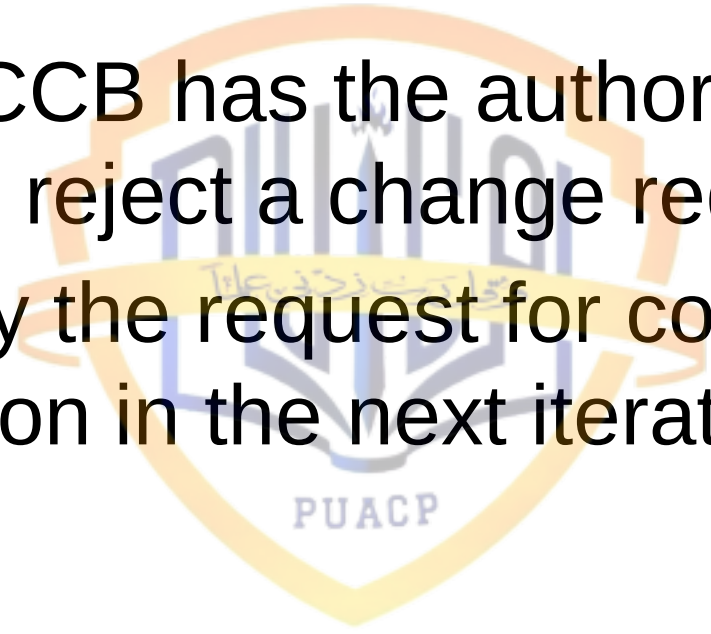
Change Control Authority/Board

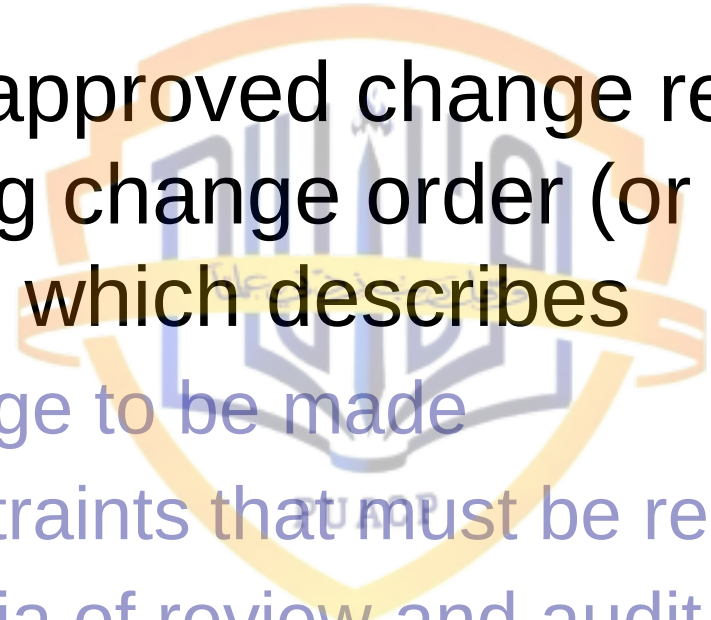
- This authority is known as change control authority (or CCA) or change control board (or CCB)
- We'll use these two terms interchangeably during this course

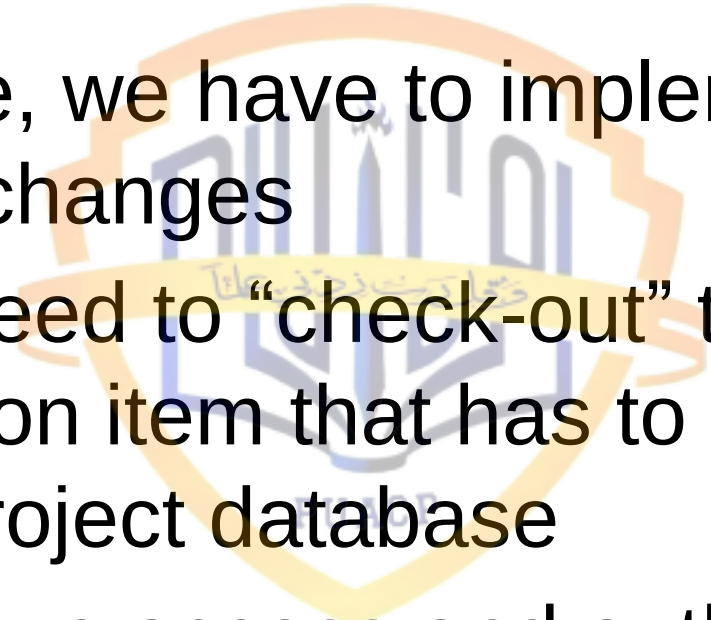
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- A CCA/CCB plays an active role in the project level change control and formal change control activities
 - Typically, a CCA consists of representatives from software, hardware, database engineering, support, and marketing, etc., depending on the size and nature of the project

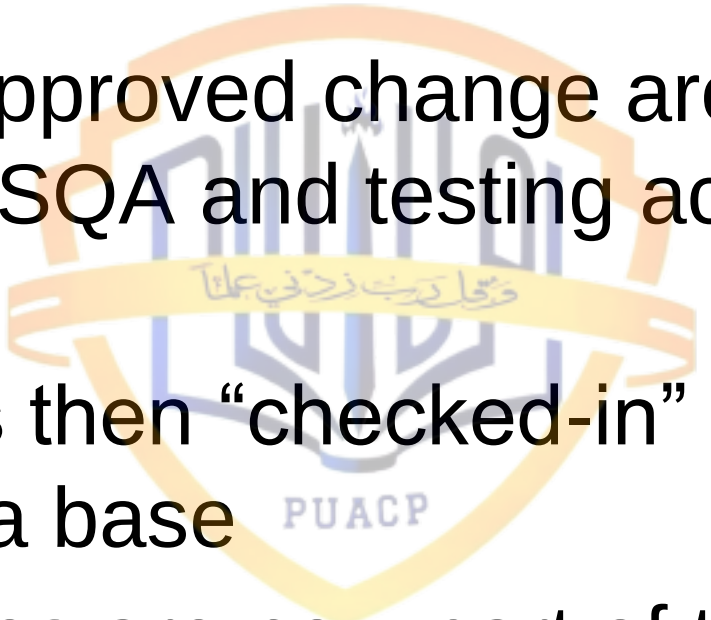
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- For every change request, the change control authority/board assesses the
 - Technical merit
 - Potential side effects
 - Overall impact on other configuration items and system functions
 - Projected cost of the change

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- In addition, change control authority/board may assess the impact of change beyond the SCI in question
 - How will the change affect hardware?
 - How will the change affect performance?
 - How will the change modify customer's perception of the product?
 - How will the change affect product quality and reliability?

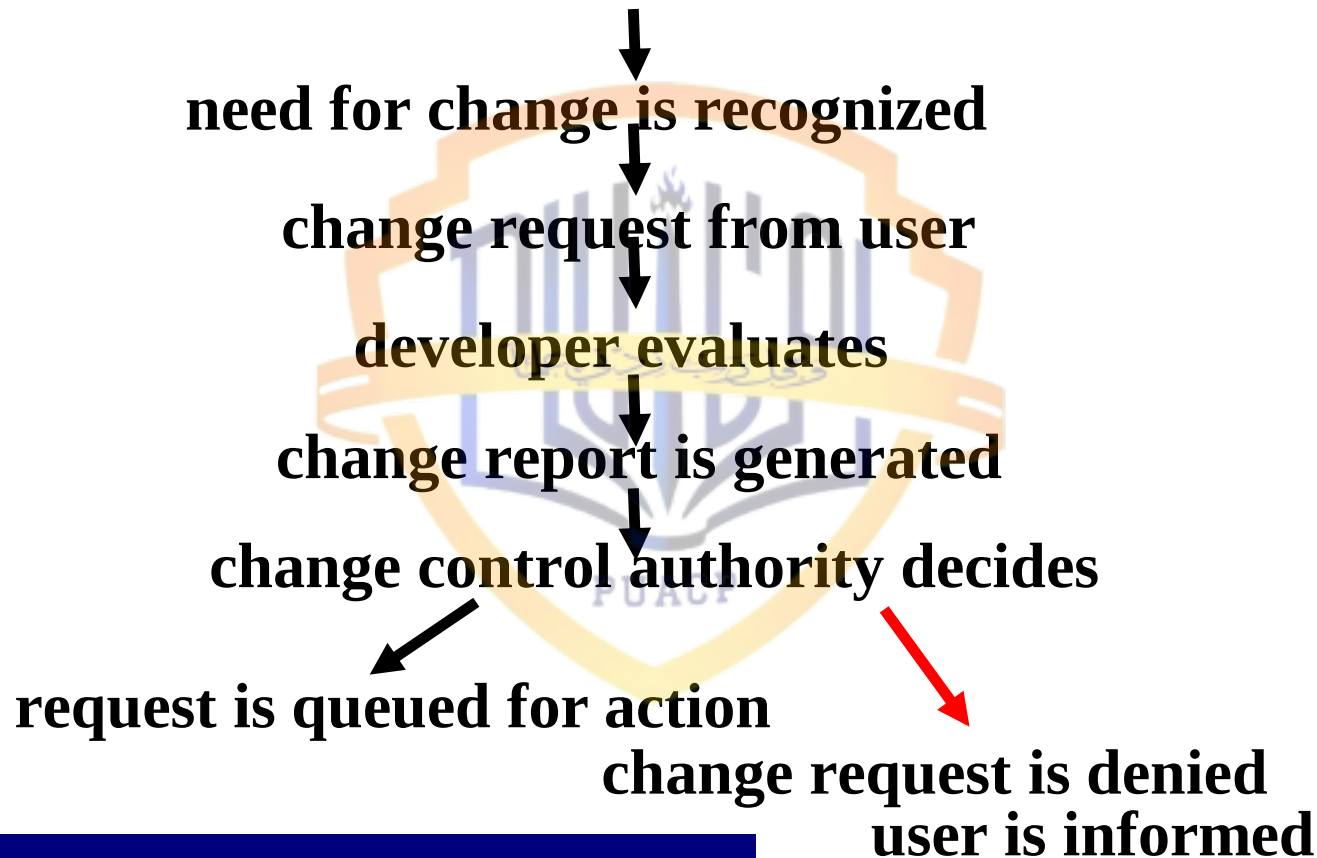
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- The CCA/CCB has the authority to approve or reject a change request
 - It can delay the request for consideration and inclusion in the next iteration also

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- For every approved change request, an engineering change order (or ECO) is generated, which describes
 - The change to be made
 - The constraints that must be respected
 - The criteria of review and audit

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- At this time, we have to implement the approved changes
 - First, we need to “check-out” the configuration item that has to be changed from the project database
 - Now we have access and authorization to make modifications to that specific SCI

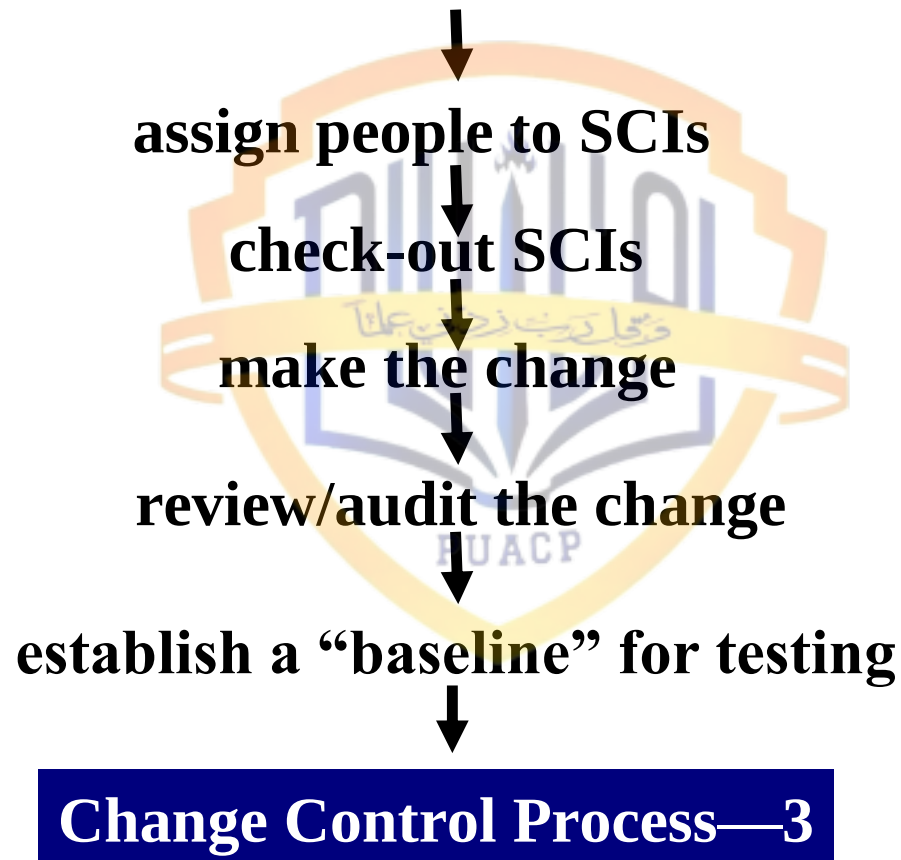
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- Then the approved change are made and necessary SQA and testing activities are applied
 - That SCI is then “checked-in” to the project data base
 - The changes are now part of the new version of the base-lined work-product

The Change Control Process - 1

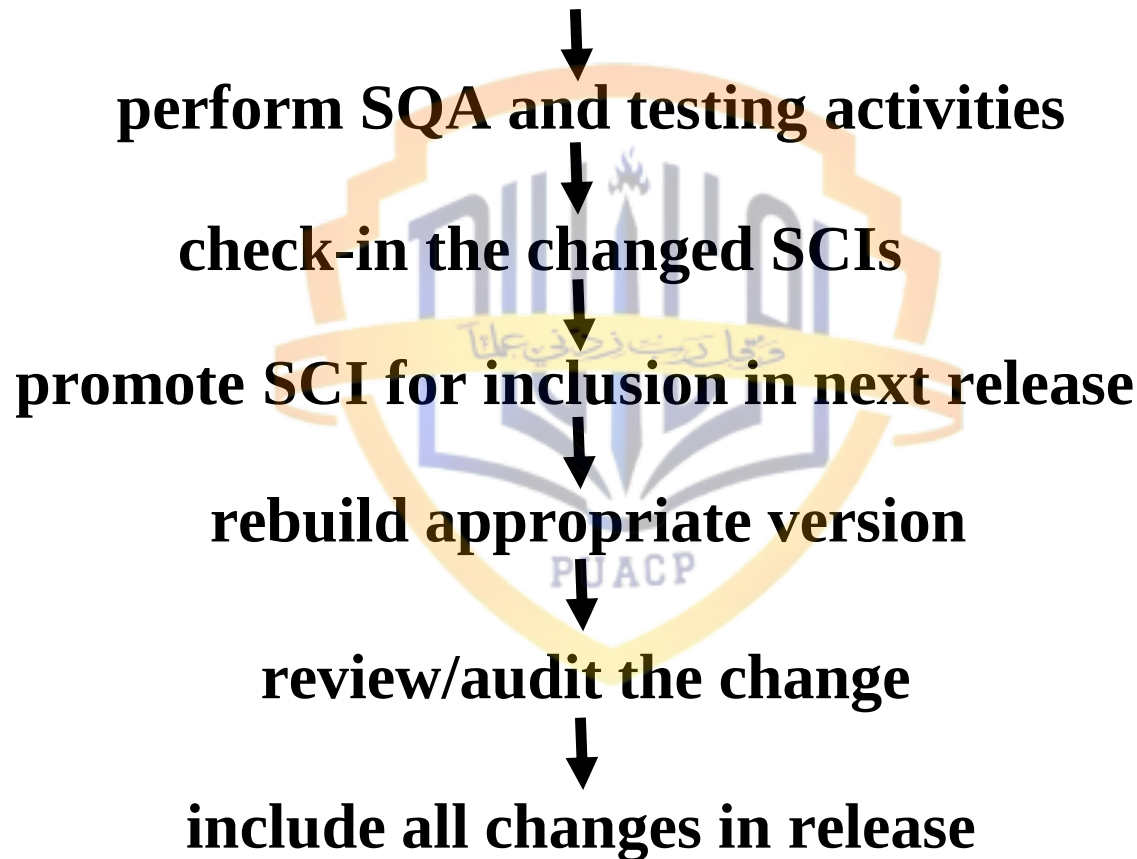


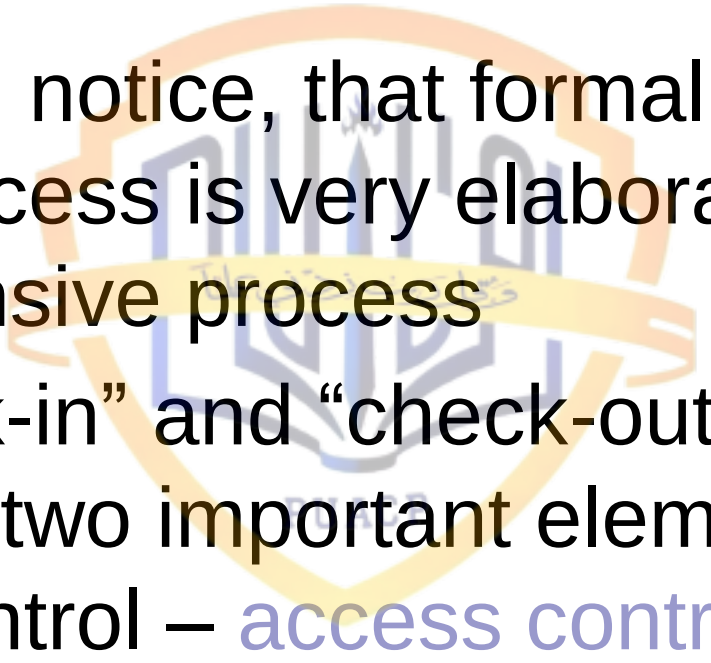
Change Control Process—2

The Change Control Process - 2



The Change Control Process - 3

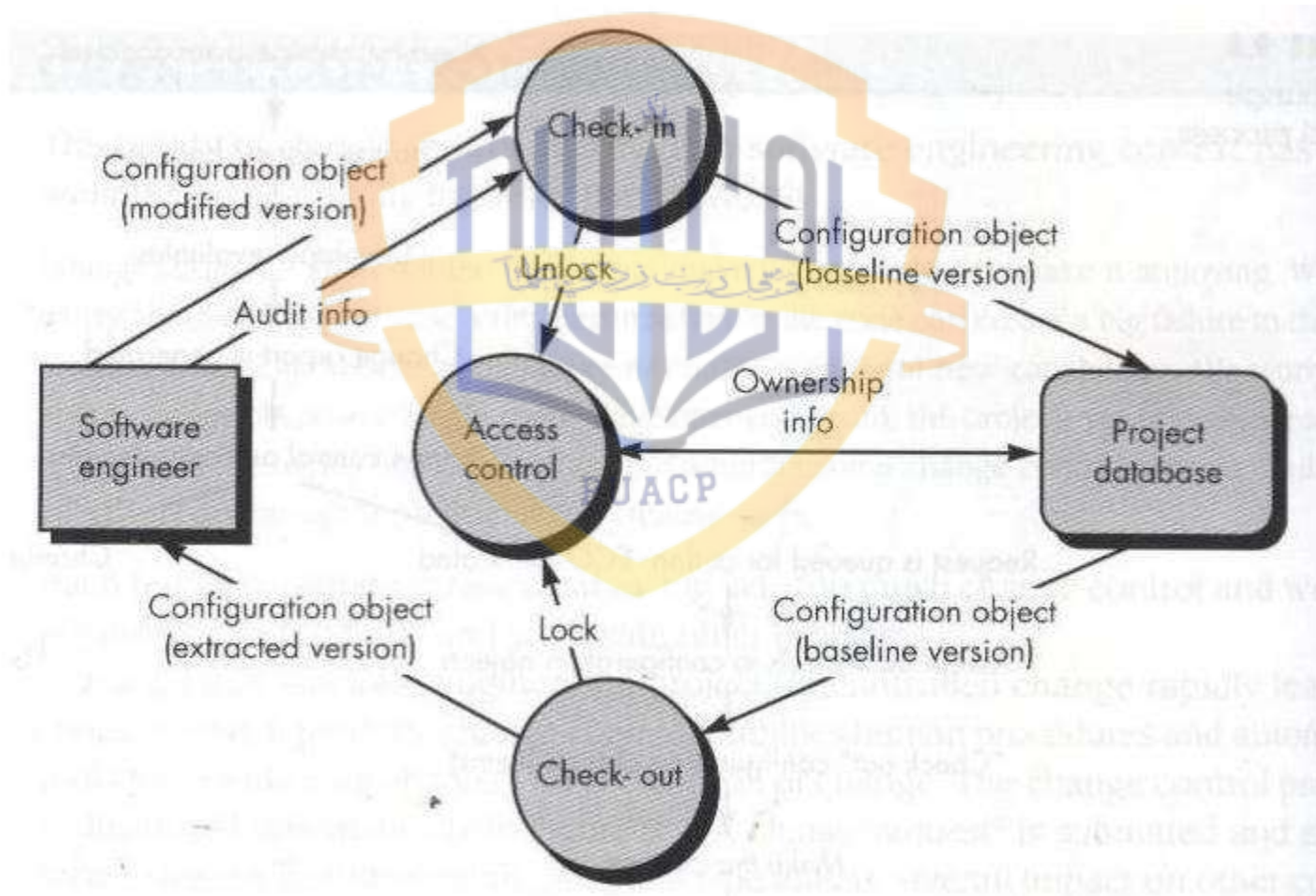


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- As you can notice, that formal change control process is very elaborate and comprehensive process
 - The “check-in” and “check-out” activities implement two important elements of change control – access control and synchronization control

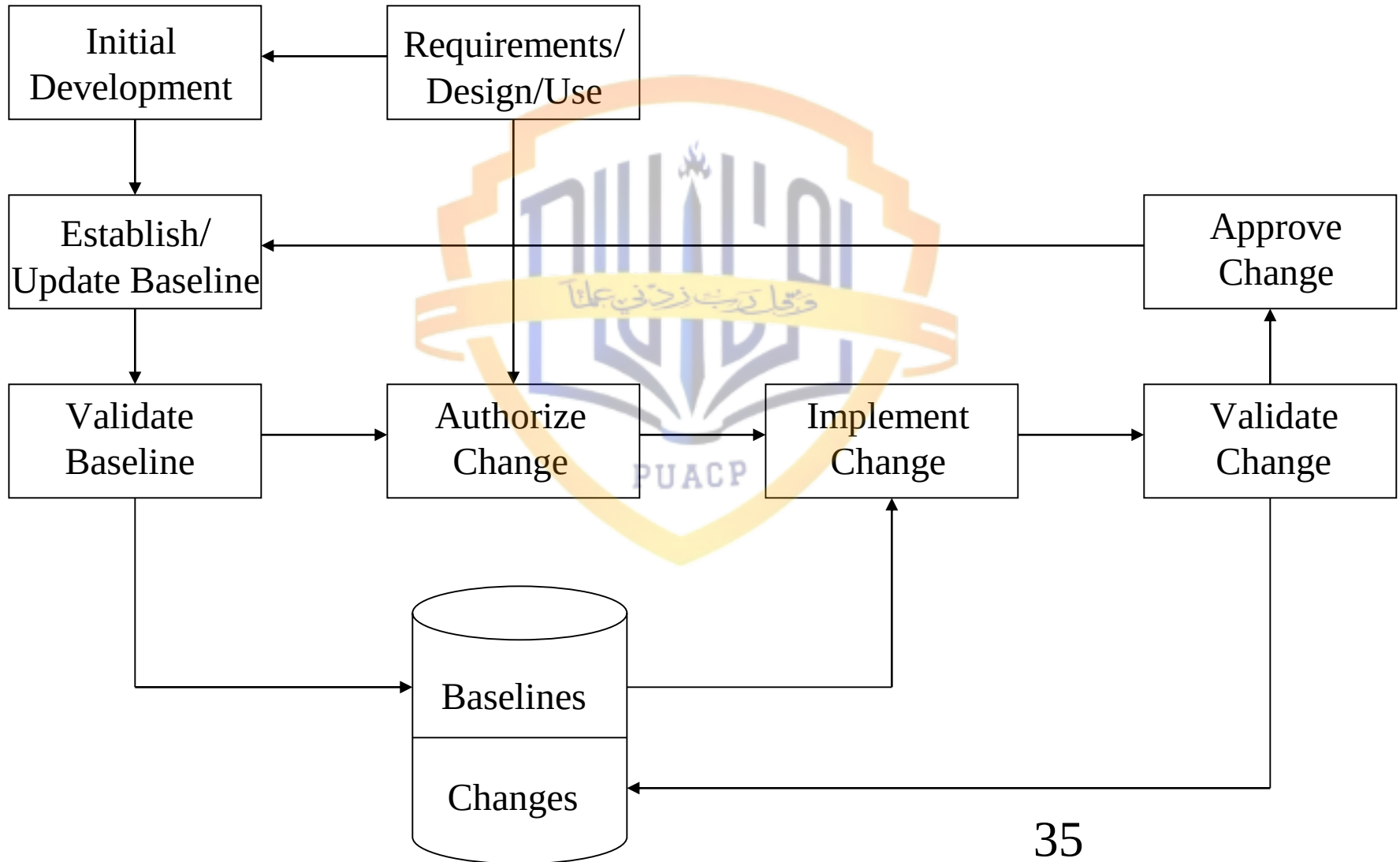
Access and Synchronization Control

- Access control governs which software engineers have the authority to access and modify a particular configuration item
- Synchronization control helps to ensure that parallel changes, performed by two different people, don't overwrite one another
- We need to implement both

Access and Synchronization Control



Software Configuration Management Overview



SCM Standards

- MIL-STD-483
- DOD-STD-480A
- MIL-STD-1521A
- ANSI/IEEE Std. No. 828-1983
- ANSI/IEEE Std. No. 1042-1987
- ANSI/IEEE Std. No. 1028-1988

Summary

- Software configuration management is an umbrella activity that is applied throughout the software process
- SCM identifies, controls, audits, and reports modifications that invariably occur while software is being developed and after it has been released
- SCM saves a project from total chaos

References

- Software Engineering: A Practitioner's Approach by Roger S. Pressman (Chapter 9.3 – 9.9)
- A Handbook of Software Quality Assurance edited by G. Gordon Schulmeyer and James L. McManus, PH, 1999, (Chapter 10.1-10.2)
- Managing the Software Process by Watts S. Humphrey, AW, 1989, page 177 (figure 7.2 SCM Overview)